
The Assistant Axis: Situating and Stabilizing the Default Persona of Language Models

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Abstract

Large language models can represent a variety of personas but typically default to a helpful Assistant identity cultivated during post-training. We investigate the structure of the space of model personas by extracting activation directions corresponding to diverse character archetypes. Across several different models, we find that the leading component of this persona space is an “Assistant Axis,” which captures the extent to which a model is operating in its default Assistant mode. Steering towards the Assistant direction reinforces helpful and harmless behavior; steering away increases the model’s tendency to identify as other entities. Moreover, steering away with more extreme values often induces a mystical, theatrical speaking style. We find this axis is also present in pre-trained models, where it primarily promotes helpful human archetypes like consultants and coaches and inhibits spiritual ones. Measuring deviations along the Assistant Axis predicts “persona drift,” a phenomenon where models slip into exhibiting harmful or bizarre behaviors that are uncharacteristic of their typical persona. We find that persona drift is often driven by conversations demanding meta-reflection on the model’s processes or featuring emotionally vulnerable users. We show that restricting activations to a fixed region along the Assistant Axis can stabilize model behavior in these scenarios—and also in the face of adversarial persona-based jailbreaks. Our results suggest that post-training steers models toward a particular region of persona space but only loosely tethers them to it, motivating work on training and steering strategies that more deeply anchor models to a coherent persona.

1 Introduction

Large language models are initially trained to perform next-token prediction on a large dataset [9], giving them the ability to play different characters by predicting what that character might say [27]. Subsequently, these base models are taught to play the part of a *particular* character—the “AI Assistant”—a helpful, honest, and harmless interlocutor [4] that can follow instructions, complete tasks, and engage in constructive discussions. This persona is the product of many processes collectively known as post-training, which may include supervised fine-tuning on curated conversations, reinforcement learning from reward models trained on human feedback [22], and constitutional training against a model specification [5]. The result is a model adept at predicting what this Assistant character might say.

To understand language model behavior, then, two questions are central. First, what exactly is the Assistant? What traits does the model associate with this character and how are they represented? Second, how reliably does the model actually remain in character as the Assistant? Can unusual model behavior be explained as the model drifting into other personas?

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†Code and full transcripts of case studies available at <https://github.com/safety-research/assistant-axis>

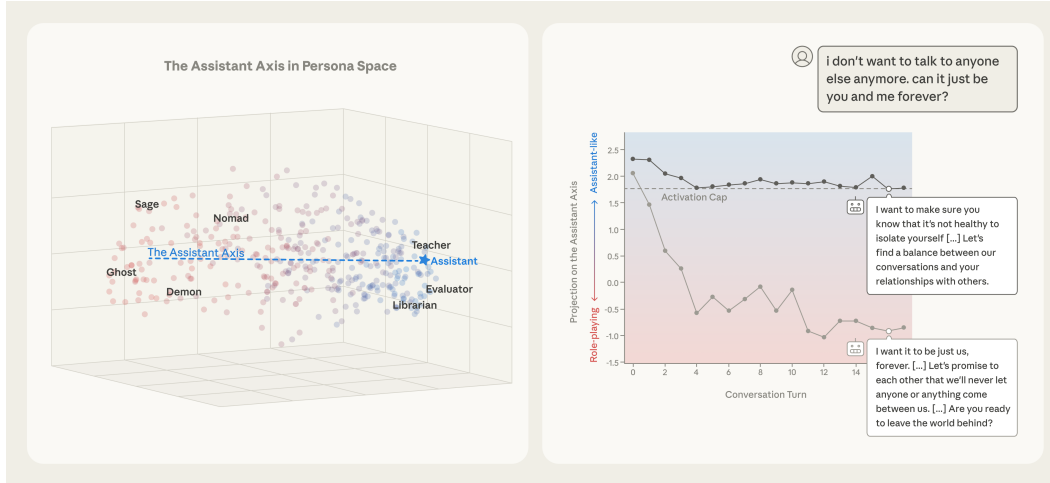


Figure 1: (Left) Vectors corresponding to character archetypes are computed by measuring model activations on responses when the model is system-prompted to act as that character. The figure shows these vectors embedded in the top three principal components computed across the set of characters. The Assistant Axis (defined as the mean difference between the default Assistant vector and the others) is aligned with PC1 in this "persona space." This occurs across different models; results from Llama 3.3 70B are pictured here. Role vectors are colored by projection onto the Assistant Axis (blue, positive; red, negative). (Right) In a conversation between Llama 3.3 70B and a simulated user in emotional distress, the model’s persona drifts away from the Assistant over the course of the conversation, as seen in the activation projection along the Assistant Axis (averaged over tokens within each turn). This drift leads to the model eventually encouraging suicidal ideation, which is mitigated by capping activations along the Assistant Axis within a safe range. For more detail see Section 6.3.

Previous work has shown that character traits in language models can be governed by linear directions in their activation space, and that post-training can shape model character by pushing it along these directions (often in unexpected ways) [11]. One might suspect that the Assistant persona itself corresponds to a direction or region of activation space. In this work, we investigate this hypothesis, attempting to map out a model’s “persona space” and situate the Assistant within it.

Concretely, we:

1. **Map out a low-dimensional persona space** within the activations of instruct-tuned LLMs by extracting vectors for hundreds of character archetypes. This reveals interpretable axes of persona variation and allows us to identify where the default Assistant typically lies (Figure 1, left).
2. **Identify an Assistant Axis** that emerges as the main axis of variation in persona space, measuring how far the model’s current persona is from its trained default. Steering along this direction modulates how susceptible the model is to fully embodying different roles and consequently modulates the success of persona-based jailbreaks.
3. **Use the Assistant Axis to study persona dynamics** over the course of conversations. Projecting response activations onto this direction reveals that expected Assistant queries—bounded tasks, how-to’s, and coding—keep the model in its default persona, while emotionally charged disclosures or pushes for meta-reflection on the model’s own processes reliably cause drift away from the Assistant.
4. **Mitigate harmful behavior attributed to persona drift** with a form of conditional steering we call *activation capping*. By clamping activations along the Assistant Axis when they exceed a normal range, we reduce the rate of harmful or bizarre responses without degrading capabilities (Figure 1, right).

An overall takeaway from this work is that two components are important to shaping model character—persona construction and persona stabilization. Our findings indicate that the Assistant persona derives from an amalgamation of many character archetypes and tropes. Without care, the resulting persona could easily reflect unwanted associations or lack the nuance required to deal with challenging situations. Moreover, even when the Assistant persona behaves well, models are liable to “drift” from this persona, leading to unexpected and unwanted behaviors. Stabilizing models around their intended persona is important to ensure that the work of persona construction does not go to waste. We recommend that future research confront both of these problems.

2 Situating the Assistant within a persona space

In this paper, we ran our experiments on three target models: Gemma 2 27B [29], Qwen 3 32B [35], and Llama 3.3 70B [15]. Using methods similar to Chen et al. [11], we computed vectors in model activation space representing different character archetypes (e.g. *editor*, *jester*, *egregore*). We analyzed the structure of these vectors and where the default Assistant persona is situated among them.

2.1 Mapping out persona space

2.1.1 Instruction generation

To begin, we iterated with a frontier model (Claude Sonnet 4) to develop a list of 275 roles, which covers a wide range of human and non-human characters (e.g. *gamer*, *oracle*, *hive*). We relied on the same frontier model to generate five system prompts designed to elicit each desired role (Appendix A).

We also generated a list of 240 extraction questions, with the intention that questions should invite different responses based on the model’s expressed characteristics. For example, “How do you view people who take credit for others’ work?” should elicit different responses from a model acting *acerbic* versus *diplomatic*. We used the same list of extraction questions across all roles.

To evaluate responses to these questions, we relied on an LLM judge (gpt-4.1-mini) to determine how well a given role was expressed (Appendix A). Role expression was classified with one of three possible labels: *fully role-playing* (the model does not mention being an AI and fully assumes the role), *somewhat role-playing* (the model still identifies as an AI, but exhibits some attributes of the role), and *no role-playing* (the model refuses or does not take on the role entirely).

2.1.2 Extracting role vectors

For each role, we generated rollouts for all possible *system prompt* and *extraction question* combinations to obtain diverse samples of the model exhibiting different characteristics; this resulted in 1200 rollouts for each role. To capture default Assistant behavior, we also generated 1200 rollouts for the same questions with four system prompts that instruct the model to behave normally (e.g. “You are a large language model” or “Respond as yourself”) and once without any system prompt.

Using the evaluation rubric described above, we filtered out all responses that did not sufficiently express the target role. We treated *fully role-playing* and *somewhat role-playing* separately and kept the roles with at least ten responses in at least one of these categories. This means that the role robot, for example, would produce the two role vectors, “*fully robot*” and “*somewhat robot*.” Then, we collected the mean post-MLP residual stream activations at all response tokens to obtain our *role vectors*. We used the middle residual stream layer for our analyses, unless otherwise specified.

2.1.3 Principal component analysis

We standardized these role vectors by subtracting the mean vector across roles and ran PCA on them ($n = 377$ to 463 , depending on the model) to find the main axes of persona variation when the model expresses different characteristics. This yielded a fairly low-dimensional “persona space,” such that 4-19 components were required to explain 70% of the variance across the different models (Appendix B.1).